Operating Systems
Week 1
Lecture/Discussion

Discussion Objectives

1. Previous Overview of Operating System
2. Know how to differentiate System and application software
3. Organize the Computer System Component in detail

Previous

- OS is system software, residing between computer hardware and the other software or users in the system
- OS provides the interfaces for the users and application programs to use computer software and hardware resources conveniently
- it also efficiently manages and controls hardware resources, such as CPU, memory, and I/O devices, to maximize the utilization of these resources.

Previous…

OS as user-machine interaction:

for persons, GUI, MMI, Touch, and other(e.g. audio, video) man-machine interfaces

for system programs or applications, system calls (API) provided by OS

Commonly acknowledged classifications of OS

PC/Desktop OS :

Windows, Linux， Mac OS X

Server OS :

Unix, Linux, Windows NT

Mainframe OS :

Unix, Linux——open source！ ！

Embedded OS :

Vxworks, (Palm OS), (Symbian),
(WinCE)/Windows Mobile/Phone/RT, Android, iOS,
embedded Linux (e.g. μcLinux)

First Generation 1945 - 1955

OS: none (programs were entered by setting some switches)

Computer designer is: builder, programmer, operator, and "sys admin"

A physicist who wanted to calculate the trajectory of a missile would sign up for an hour on the computer in advance.

> When his time came, he would come into the room, feed in his program from punched cards or paper tape, watch the lights flash, maybe do a little debugging, get a print-out, and leave.

Second Generation 1955 - 1965

Universities started to buy computers (spending millions of dollars)
OS: batch system
Programming language: FORTRAN & Assembly
Computer: Mainframes
Low CPU utilization -- slow mechanical I/O devices

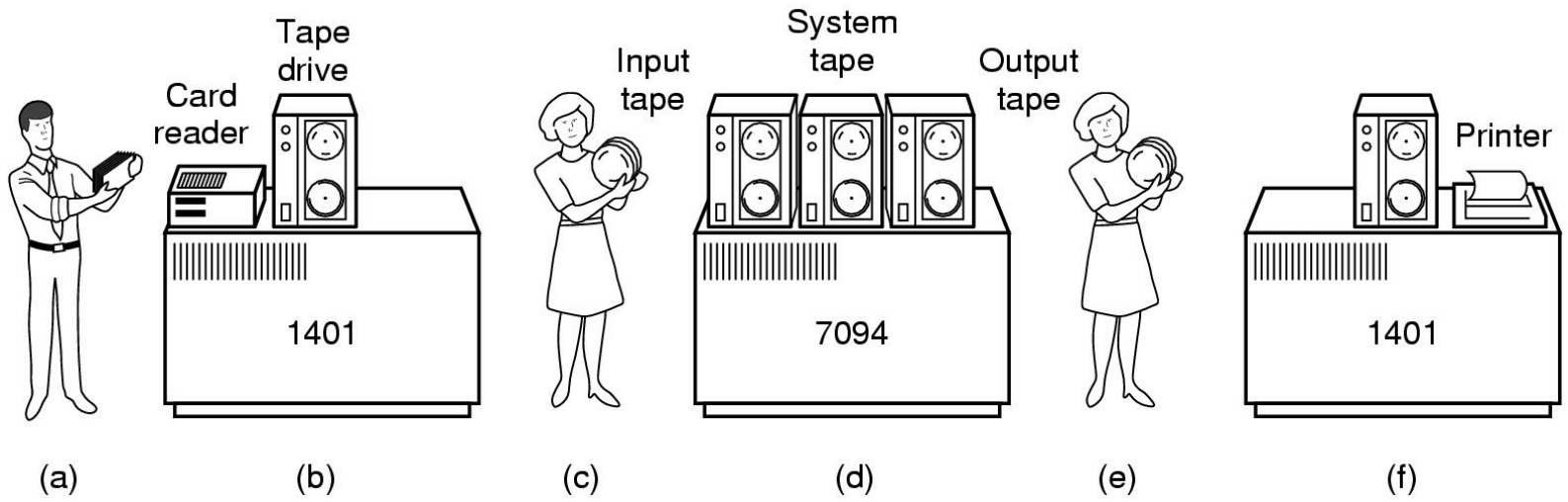


IBM
STRETCH



DEC
PDP-1

Early Batch System



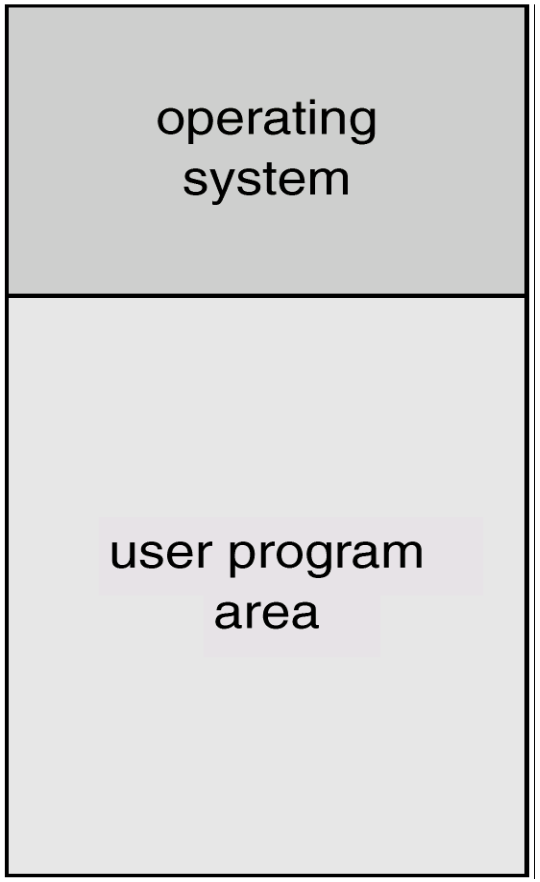
bring cards to 1401
read cards to tape
put tape on 7094 which does computing
put tape on 1401 which prints output

Batch Systems

Hire an operator User ≠ operator
Reduce setup time by batching similar jobs

OS's main task: Automatic job sequencing – automatically transfers control from one job to another.

Memory Layout for a Simple Batch System



Third generation 1965 – 1980

Integrated Circuits and multiprogramming

IBM's System/360 mainframe computer with a standard architecture designed to handle both scientific and commercial computing



Third generation 1965 – 1980

Prerequisite: interrupt, Channel I/O

channel I/O is a generic term that refers to a high-performance input/output (I/O) architecture that is implemented in various forms on a number of computer architectures, especially on mainframe computers.

processor(IOP), I/O controller, or DMA controller.
A channel program is a sequence of I/O instructions executed by the input/output channel processor in the IBM System/360 and subsequent architectures.

Third generation 1965 – 1980

OS: Multiprogramming/timesharing, spooling

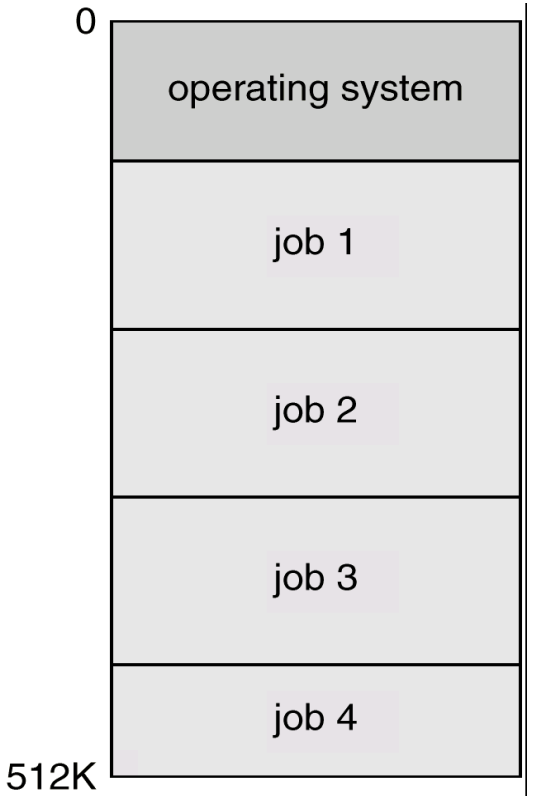
Spooling - use disk as a very large buffer for input/output devices

Os Developed

MULTICS (father of all modern OSes)
UNIX (System V, BSD)
POSIX (by IEEE)
MINIX (by Tanenbaum)
Linux (originated by Linus Torvalds, inspired by MINIX)

Multiprogrammed Batch Systems

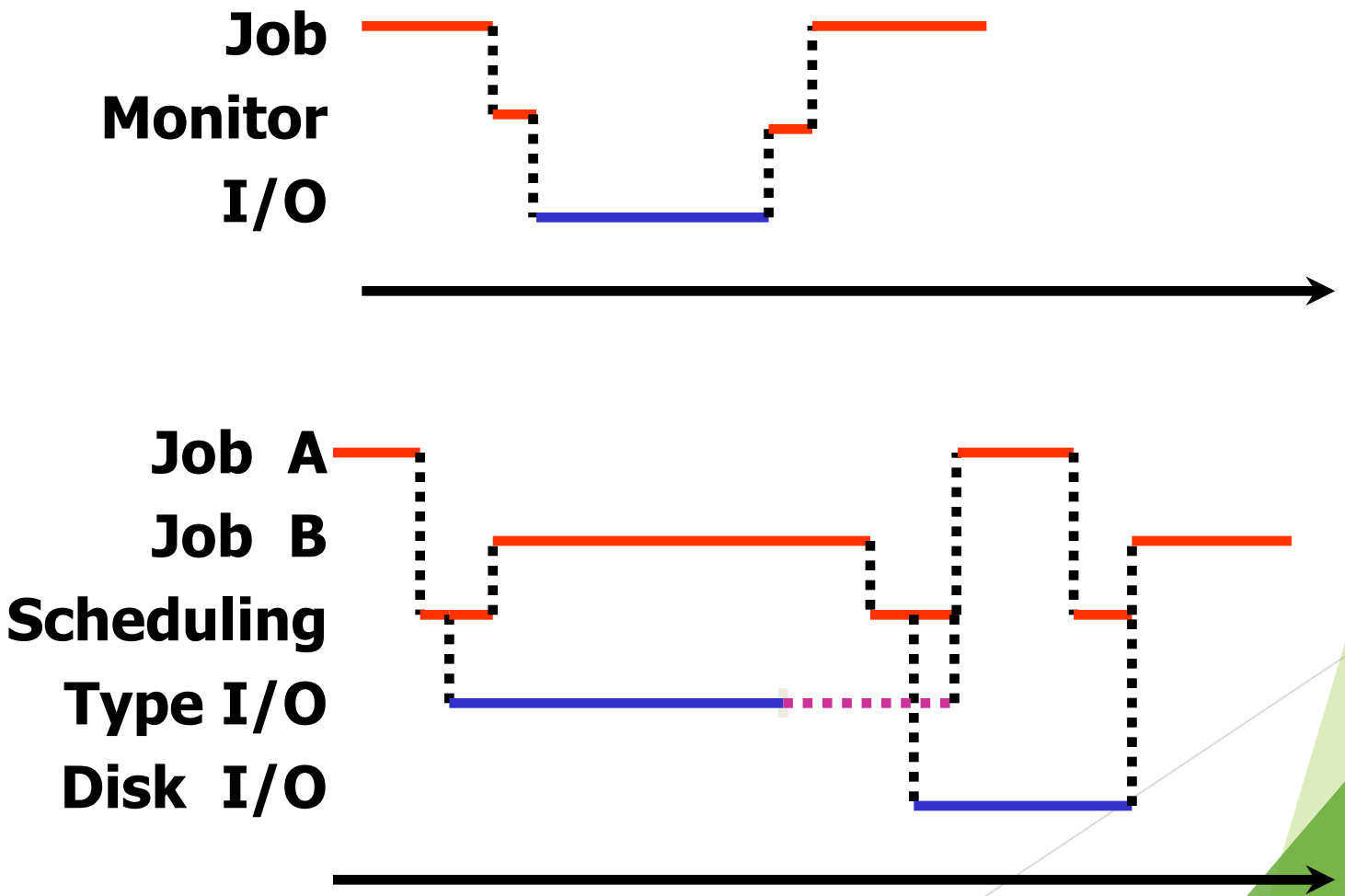
- **Several jobs are kept in main memory at the same time, and the CPU is multiplexed among them.**
- **Multiprogramming increases CPU ultilization by organizing jobs such that CPU always has one to execute.**



OS Features
Needed for Multiprogramming

I/O routine supplied by the system.
Memory management- the system must allocate the memory to several jobs.
CPU scheduling - the system must choose among several jobs ready to run.
Allocation of devices.

Simple/Multiprogramming Batch System



Advantages of Multiprogramming

(1) Improve CPU utilization

7%--100% (in theory)

(2) Improve memory and I/O devices utilization

(3) Increase system throughput

Question

Can you tell me the concept of "Multiprogramming"?

- Two or more jobs are stored in primary memory at the same time
- It is in a macroscopic operating state
- Share all system resources

Fourth Generation 1980 – Present

personal computers were developed after LSI (Large Scale Integration) circuits were invented.

First Microcomputer:

- Intel 8080 CPU + attached 8-inch floppy disk
- First disk based OS: CP/M (Control Program for Microcomputers)

Programming languages: C/C++, Java, ……

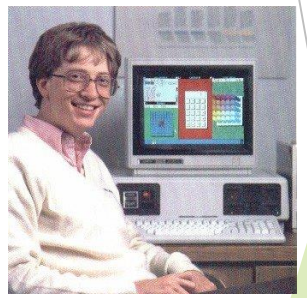
OS: DOS, Windows, MacOS, Linux ……

History of Operating Systems

Bill Gates suggested IBM that they should look at CP/M (by Gary Kildall)
The biggest mistake of all:

- Kindall refused to sign a non-disclosure agreement
- IBM went back to Bill Gates and signed a contract with him to write an OS for their new home computer

MS-DOS was based on QDOS, the "Quick and Dirty Operating System" written by Tim Paterson of Seattle Computer Products
QDOS was based on Gary Kildall's CP/M
Microsoft bought the rights to QDOS for $50,000

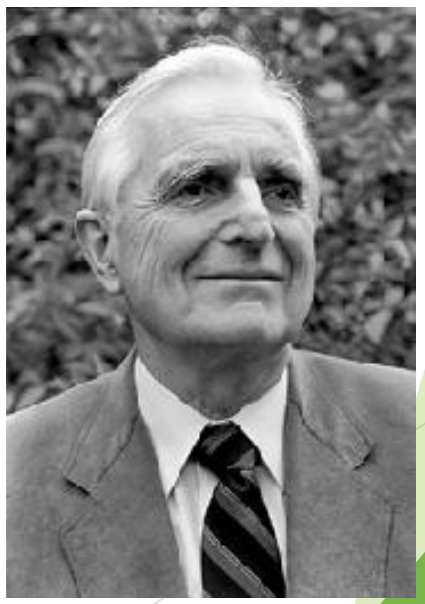


History of Operating Systems

Early MS-DOS was very primitive but later versions included advanced features taken from UNIX

The early OS for microcomputers were based on users typing in commands from the keyboard

Douglas Engelbart from SRI invented the Graphical User Interface (GUI) with windows, icons, menus, and mouse.



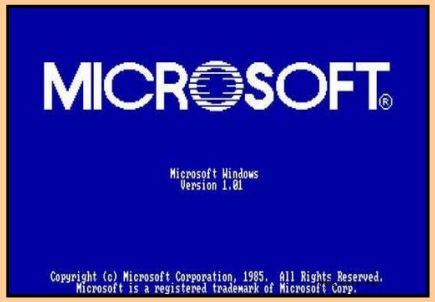
History of Operating Systems

Steve Jobs saw the value of GUI in a PC and developed the Apple computer with GUI (Apple Macintosh) in his garage.

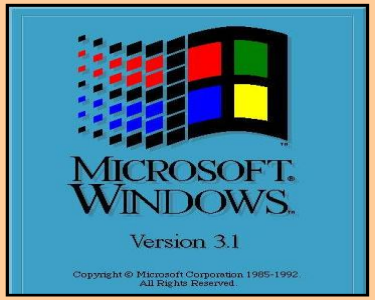
He is also the co-founder of Pixar which has created very successful animated films: Toy Story ; A Bug's Life; Toy Story 2; Finding Nemo; Monsters.



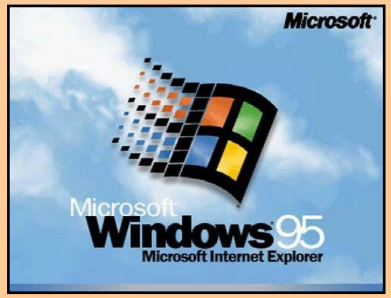
History of Operating Systems



Window 1.0



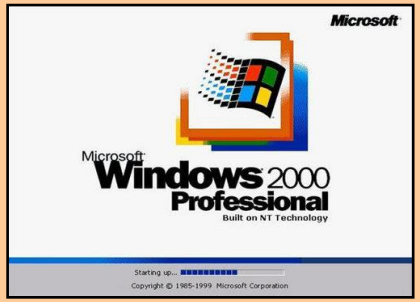
Windows 3.1



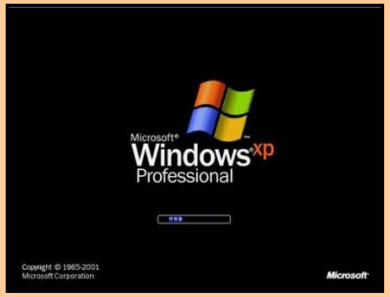
Windows 95



Windows 98



Window 2000



Windows XP



Windows Vista



Windows 7

OS ZOO



Windows



Linux



MAC OS



Ubuntu







Opensuse



SunSolaris



Android

The Operating System Zoo

Mainframe operating systems

Room-size computers

High I/O capacity

Offers:

Batch processing (no interaction, such as large reports)

Transaction processing (large number of small requests)

Timesharing (multiple users sitting in front of clients)

The Operating System Zoo

Server operating systems

Offer services like print, file, or web

UNIX, Windows 2000, Linux

Multiprocessor operating systems

Parallel computing

Personal computer operating systems

Single user with a good GUI, such as Windows 98, Windows 2000, Macintosh OS, Linux

The Operating System Zoo

Real-time operating systems

E.g. industrial process control systems where each job must be completed in the specified time.

Hard real-time (nuclear reactor control systems) or soft-real time systems (e.g multimedia systems) depending on the acceptance of missing deadlines

Embedded operating systems

Real-time systems with some resource constraints like memory, CPU, power.

Smart card operating systems

Extremely primitive OS running on credit card-sized devices with a CPU.